

STAC Users Guide

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A Roland Smart Tally Client

Take a bit of custom software, pour it into an [M5Stack ATOM Matrix](#) board, give it a shake and—ta da! It becomes a *Smart Tally Atom Client* or **STAC** for short.

Its purpose and sole joy in life is to monitor the tally status of a single video input channel of a Roland video device that implements their Smart Tally protocol and indicate to you that status via the display of the ATOM.

The STAC uses WiFi to connect to the same network as that of the Roland switch. It continually polls the switch for the tally status of the video channel being monitored. For that channel, the STAC will set the colour of its display:

- When in "Camera Operator" mode, to:
 - RED if the channel is in PGM (Program or onair)
 - GREEN if the channel is in PVW (Preview or selected)
 - "PURPLE DOTTED" if the channel is not in either PGM or PVW (unselected)
- When in "Talent" mode, to:
 - RED if the channel is in PGM (Program or onair)
 - GREEN otherwise

There is also a Peripheral Mode that allows a STAC to be connected to another via their GROVE ports. The STAC configured in Peripheral Mode receives its data from the other STAC without the need to establish a WiFi connection to the Roland switch.

Setup of the STAC for the WiFi credentials, Roland switch IP address, port number, number of tally channels, polling interval and performing maintenance tasks is done using a web browser.

Other parameters are set using the display button, with feedback provided via the display.

To ensure you see the pictures, make sure the `images` folder is in the same folder as this

Notes

1. Starting with version 3 of the STAC software, many more hardware devices are supported. This guide uses the M5Stack ATOM Matrix when presenting features and examples. Other devices operate in much the same way though there are differences in things like button locations and, for devices with LCD displays, how things look.
2. It is also assumed the ATOM Matrix has already been programmed ("flashed") with STAC software version 3 or greater. If that's not the case:
 - Download the firmware from the [STAC Releases page](#)
 - See [bin/BIN_README.md](#) for flashing instructions
 - Then return here to continue setup

Device-specific supplements can be found in the [bin folder](#) - look for your device's folder (e.g., `ATOM_Matrix`, `StickC_Plus`, etc.).

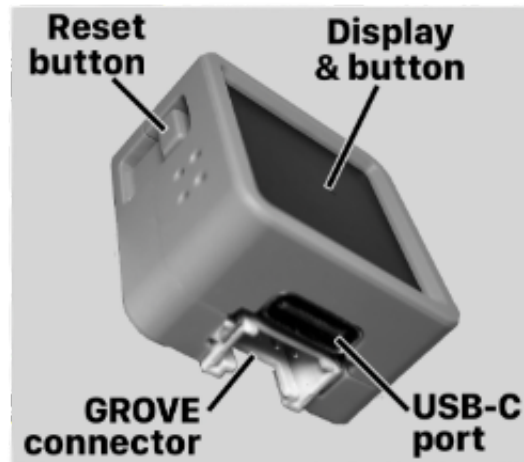
Contents

- [STAC Users Guide](#)
 - [Contents](#)
 - [Getting to Know You](#)
 - [Quick Start](#)
 - [First Time Setup](#)
 - [Configuring Multiple Devices](#)
 - [Up and Running](#)
 - [Setting the Tally Channel \(V-60HD\)](#)
 - [Setting the Tally Channel \(V-160HD\)](#)
 - [Setting the Tally Display Mode](#)
 - [Setting the Startup Mode](#)
 - [Setting the Brightness Level](#)
 - [Setting the Display Orientation](#)
 - [Peripheral Mode](#)
 - [Enabling Peripheral Mode](#)
 - [Operation](#)

- [Setting the Brightness Level](#)
- [Setting the Tally Display Mode](#)
- [Disabling Peripheral Mode](#)
- [Hardware Implementation](#)
 - [Connection](#)
 - [In Peripheral Mode](#)
 - [In Normal Mode](#)
- [Backwards Compatibility](#)
- [Settings Deep Dive](#)
- [Maintenance Functions](#)
- [Getting to the Maintenance Functions](#)
- [Redoing the Setup](#)
- [Updating the Software](#)
 - [Other things to know about updating](#)
- [Factory Reset](#)
 - [Using a Browser](#)
 - [From the STAC](#)
- [Deep Secrets](#)
- [Back to the Beginning](#)
- [Special Configurations](#)
- [Troubleshooting](#)
- [The WiFi Connect Loop of Death](#)
- [Tally Status not Changing](#)
- [A Big Orange X?](#)
- [A Big Purple X?](#)
- [The Red WiFi Interrupt Screen!](#)
- [Purple Question Mark?](#)
- [The Big Red X](#)
- [Acknowledgements](#)
- [LICENSE](#)
- [Revision History](#)

Getting to Know You

The bits you need to know about when using the STAC are shown in the picture below.



The Display is, well, just that. A square matrix of colour LED's used to show: the state of the tally channel being monitored; receive feedback on its operation; and for setup. We'll get to those parts in a while.

The Display is also a button that you'll click and press to select and set various operating parameters for the STAC.

The Reset Button is used in conjunction with the Display button to reconfigure, clear, and update the STAC. It can also be used to restart the STAC.

The USB port is used to power the STAC. Use one end of a standard USB-C cable to connect here and the other end to whatever matches the power supply being used. FWIW, data is not required, only 5 VDC power. There is no power switch on the STAC so once the cable is connected to a power supply it comes to life and is good to go. The power can be disconnected at any time without concern for the state of the STAC.

The GROVE connector is used when the STAC is operated in Peripheral Mode. Connecting a cable between two STACs allows the peripheral STAC to be powered by, and receive tally status information from, another STAC. There's a [section](#) on this later.

On the back of the STAC is a dang tiny 2mm machine screw hole that can be used to attach the STAC to whatever you would like.

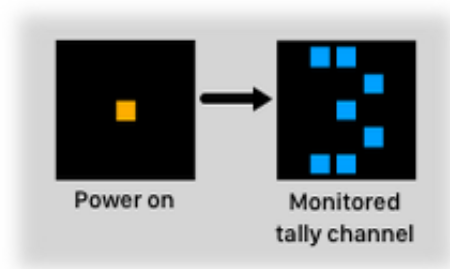
Note the ventilation holes on the sides of the STAC. To allow for cooling, these should not be blocked. The display LED's can get a bit toasty, especially when operating at higher brightness levels.

Quick Start

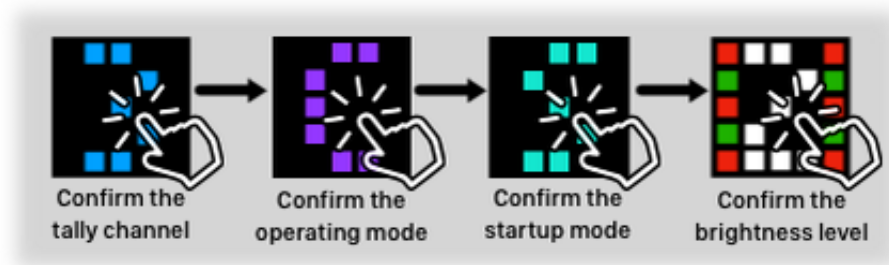
We'll get to the initial setup (and how to change it) later. Here we'll show the normal operating sequence for using the STAC.

Connect one end of a USB-C cable to the STAC and the other to a 5 VDC power source.

The STAC will briefly show an orange "power on" indicator, which will change to green briefly to let you know all is good with the STAC and then display the tally channel of the Roland Smart Tally device it is monitoring. This is the active channel.



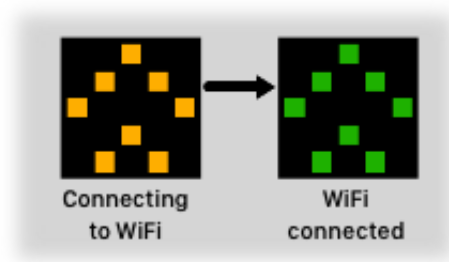
From here, click the Display Button. The Tally Display Mode of the STAC is shown. Click again. The Startup mode of the STAC is shown. Click once more and the current Brightness Level of the display is shown.



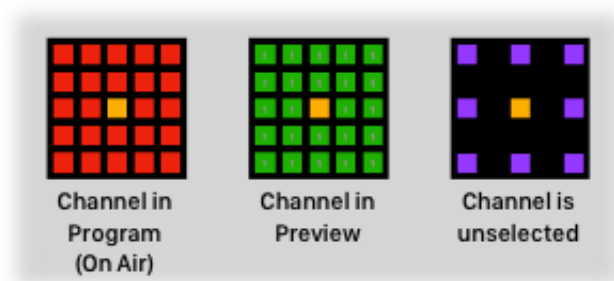
The active tally channel, tally display mode, startup mode, and the display brightness can each be changed when they are shown during startup. The display brightness level can also be changed when the STAC is actively monitoring the tally channel. We're getting to that part too.

When the Brightness Level is displayed, the next click of the display button (we'll just call this the button from here on), will let the STAC loose to do its magic.

The orange "attempting to connect to WiFi" icon is shown and, after a bit if the connection is made, the icon will change to green, indicating success.



Once WiFi is connected, the STAC will start querying the configured Roland Smart Tally device for the tally status of the channel being monitored. The STAC then continually polls the Roland switch and the display will change according to the tally status returned by the Roland device like so...



We're going to call this the normal operating state of the STAC.

Note:

- if the STAC Tally Display Mode is set to Talent, only the Program and Preview states will be shown.
- if you're seeing the four corners blinking green off and on, the STAC Autostart mode is active. We'll cover that [below](#).

So that's the general flow of things. Now let's get to the nitty gritty...



First Time Setup

Until you tell it a few things, the STAC is pretty useless. You'll have to set it up.

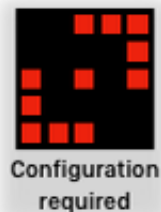
To do this you need the:

- SSID and password of the WiFi network to which your Roland switch is connected;
- IP address of the Roland switch;

- the port number used by the Roland switch;
- the desired maximum tally channel number(s) of the Roland switch;
- for the Roland V-160HD model switch, the LAN Control Password.

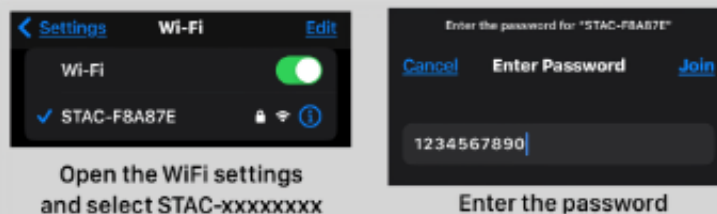
You'll also need a computer, phone, tablet, or some other device that can connect to a WiFi network and run a web browser as that is the tool you'll use to send the setup information to the STAC. Here's how...

1. Power up the STAC. It will flash the red "setup required" icon and then hold this icon on the display, pulsing in brightness to let you know it's waiting. The STAC then fires up a WiFi hotspot so it can communicate with your web browser and waits for the setup information to be received.



2. Using a computer, phone, tablet, or some other smarty pants device, open its WiFi settings and pick the SSID of your STAC. The STAC SSID starts with "**STAC-**" followed by eight characters. Something like "**STAC-4660A124**".

Each STAC has its own unique SSID.



3. Join the STAC WiFi hotspot and when prompted, enter the STAC WiFi password which is **1234567890**.
4. Once connected—on your smart device: fire up a web browser; open a new browser window and go the STAC web interface page at `http://stac.local`.
5. Make sure the Setup tab is selected, then tap the model list to pick the switch you are configuring and then tap the "Next" button.

6. Fill in the form with the: WiFi SSID and Password; the Roland device Smart Tally IP address; its port number; and the maximum channel number that you want to monitor. Remember, the Roland device must be on the WiFi network that matches this SSID and Password. The Polling interval default is OK for most uses.

For the V-160HD; the LAN password is required and the maximum tally channel number can be specified for the HDMI and SDI channels independently.

If you need to correct the model, tap the "Back" button to return to the Model Selection page.

7. Once the information has been entered, tap the "Submit" button to send the form to the STAC.

The STAC will: send a confirmation page to your browser; briefly show a confirmation checkmark on the display; shut down its WiFi hotspot; and then restart.

STAC Setup

Device: M5Stack ATOM Matrix
SSID: STAC-00F8A8
MAC: 94:B9:7E:A8:F8:00
Version: 3.0.0-RC.25 (a4f031)

Setup Maintenance

Device Setup

⚡ Quick Setup
Load configuration from a file or you can paste the settings from the clipboard.

Use settings file:

Choose File No file chosen

Paste Settings

Select Roland Switcher Model:

✓ Choose model...
V-60HD
V-160HD

WiFi Settings

Network Name (SSID):
RolandSSID

Password:

V-60HD Settings

V-60HD IP Address:
192.168.3.16

Port:
80

Max HDMI Channel (1-8):
8

Poll Interval (ms):
301

Configure STAC

Reset Back

Copy Settings Save Settings

Copy to paste into another STAC, or Save as file for backup.

Configuration Saved

Your STAC configuration has been saved successfully.
The device will now restart and connect to your WiFi network.
Please wait...

The configuration is confirmed in the browser...

and on the display.

Pick the Switch Model and click "Next"

Fill in the form and click "Configure STAC"

Configuring the STAC

8. Close the confirmation window in your browser, reconnect your smart device to your preferred WiFi network and you're done. Like dinner.

After the configuration information is submitted and the STAC confirms receipt, the STAC will start up as described in the [Quick Start](#) section above. Unless it doesn't, in which case, pop down to the [Troubleshooting](#) section and it'll tell you how to fix that up. And since you're wondering yup—you sure can redo the setup. Nothing is forever. There's a how-to [below](#).

But first, a few things about entering the setup information in your browser:

- The Model, SSID, Roland IP address, port number and maximum tally channel number are required. The form cannot be submitted with these fields blank.
- The maximum SSID length is 32 characters
- The password can be left empty to allow for open WiFi networks. The password is limited to 63 characters.
- The Roland IP address must be in proper IP address format before the form can be submitted.
- For the V-60HD, the maximum tally channel number is set to **6** by default. The highest number that can be entered is **8**. The smallest is **1**.
- For the V-160HD, the maximum and default HDMI and SDI tally channel numbers are **8** . The smallest channel number for each is **1**.
- The port number is set to a default of **80**. It should only be changed in special situations like described in [Special Configurations](#) below.
- The Polling interval is the amount of time the STAC will wait between sending tally channel status requests to the Roland device. Default is **300** ms. Minimum is 175 ms and the maximum is 2000 ms. Though the value can be set in 1 ms increments, the practical "chunk" is increments of 50 ms.
-  **WARNING:** though the form will let you enter a V-160HD Network User ID, it should be left as "user". Any other value will not let the STAC communicate with the switch.

Regarding the SSID and Password, be cautious when entering quotes as some browsers will by default convert these to "smart quotes" which, as per WiFi standards are not allowed.

If the Submit button is tapped and there is missing or incorrect information entered on the form, a little bubble will pop up by that field prompting you to complete or correct the information.

The SSID and Password fields stop accepting input after their limits are reached.

Taping the "Reset" button on the browser form will clear all information entered and set the port number, number of channels and the polling interval to their defaults, leaving the form open.

At the top of the form some info about the STAC is shown.

- Device is, well, the hunk of hardware you're connected to.
- SSID is the unique identifier for this STAC. It is the SSID of the STAC when being configured and the label you'd see in the WiFi router client table for this STAC when it is operating.
- MAC is the network interface MAC address. Skip ahead a couple of paragraphs for why you might need that.
- Version is the version number of the STAC software. The characters in brackets are the "build number" of this version.

If you're having trouble connecting to the STAC using `http://stac.local`, it may be because the `http:` bit wasn't entered and so the browser is trying to go use an `https:` connection. If checking that doesn't work, you can always connect by pointing the browser to the IP address of the STAC which is `192.168.6.14`.

One last point regarding the WiFi network. Configuring the STAC with a static IP address is not supported. Therefore, when the STAC negotiates a connection to the specified network, the network must supply an IP address via DHCP. If a static STAC IP address is desired, the STAC MAC can be used to assign the IP from the router. You'll have to consult the documentation for your router on how to do that.

Egads. That was a lot of jargon. Find a friend that knows networks if you need a hand with this.

Configuring Multiple Devices

If you are configuring multiple STACs in one go, run, do not walk to the [Settings Deep Dive](#) section below. There are more features available in the Settings tab which are pretty darn cool to help with that.

Note that it may be required to close the browser Confirmation window each time before moving onto the next one. Using a desktop or laptop computer and browser is recommend. Some browsers, notably the mobile variety, get hung up when connecting to the `stac.local` or IP address of the STAC (as each one is the same) even though the STAC and its WiFi SSID are unique.

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Up and Running

OK, you got the STAC set up, it's connected to the WiFi network (of the Roland device) and you're looking to set it to monitor a tally channel, etc. Perfect! Let's get into that...

Once set up and connected to the WiFi network, you can tell the STAC:

- which switch tally channel to monitor;
- to set the tally display to "Camera Operator" or "Talent" mode;
- which startup mode to use;
- to change the brightness level of its display.

We call these the operating parameters, or parameters for short.

To change an operating parameter you use the display button. When a parameter is displayed during startup, a single click of the display button will confirm the setting of that parameter and advance to the next one. The parameter sequence is:

- Active Tally Channel →
- Tally Display Mode →
- Startup Mode →
- Brightness Level

Clicking past the Brightness Level display starts the STAC WiFi connect sequence and then on to monitor the active tally channel.

To change the value of a parameter, you press and hold the display button while that parameter is being displayed. After about two seconds, the STAC will move into "select state". The display will change to reflect that state and to let you know you're now able to change that parameter.

Once the STAC is in a select state, release the button. Subsequent clicks of the button will advance to the next value of the parameter, looping back to the start when the end value of that parameter is reached.

Once the desired value for that parameter is shown on the display, press and hold the button again for about two seconds. The STAC will briefly display a green checkmark to confirm the change, exit select state and display the new value of the parameter in its normal (non-select state) colours.

From here, a button click will move to the next parameter that can be changed. Or if you press and hold the button, you'll once again enter the select state for that parameter. Wash, rinse, repeat.

That was a lot of words. Let's show you how this works in detail for changing the tally channel.

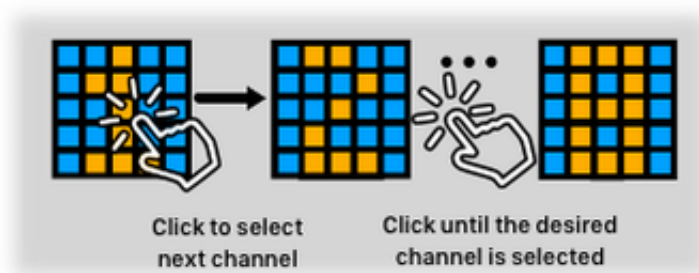
You can then apply the same method for the other parameters...

Setting the Tally Channel (V-60HD)

Power up your STAC. When you see the blue active tally number on the display, press and hold down the button. In about two seconds the display will change from a blue number on black background to an orange number on a blue background. The orange colour is a bit of a heads up to let you know the STAC is in a select state and you can now change or select the value. We use this convention with all the parameters when in a select state.



Now click the button once. The tally number will advance by one. Keep clicking the button and watch how the tally number increases and then wraps around to the number 1. Keep clicking until the desired tally channel is shown. Or until your finger gets tired. Your call.



Now press and hold the button. After about two seconds the green "confirmed" checkmark will briefly appear, the STAC will exit select state and the display will show the new active tally channel as a blue number on a black background. Release the button.



Only easy right?

When in a select state, the STAC has a timeout feature. If after entering a select state, no clicks or presses of the button are detected for about 30 seconds, the select state is cancelled and the display and tally channel will revert back to the value it had before entering select state. From here the button can be clicked to advance to the next parameter or press and hold it to enter the select state again.

Point to note. The maximum tally channel number is set when the STAC is set up via the web browser. If that maximum is set to **1**, you can enter the tally channel select state but clicking the button thereafter will have no effect as the lowest and highest numbers are the same. So if the tally number is not changing past **1**, and you need to select a higher number, you'll need to redo the setup of the STAC. No biggie, check out how to do that [below](#).

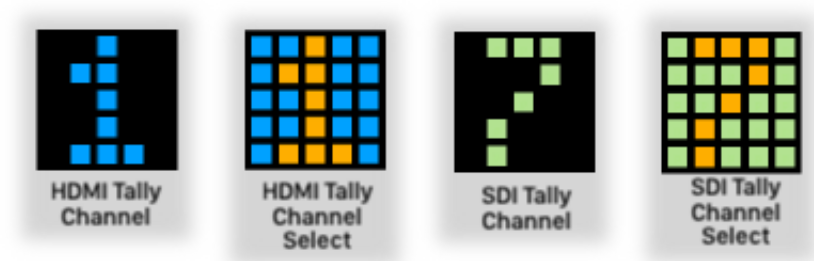
The STAC setup does not require that the maximum tally channel be the same as all the channels of the Roland device that support the Smart Tally protocol. For example, the Roland V-60HD supports up to six Smart Tally channels. However if you only have sources connected to the first four video input channels, you could configure the maximum tally channel on the STAC to be **4**. Then during setup, after selecting channel four, the STAC would wrap back around to channel **1**.

While we're talking setups, if your STAC is set to monitor, say channel **6**, and you redo the setup with the maximum tally channel set to a number less than that, after sending the new setup information to the STAC the active tally channel will be reset to **1** as the STAC can't monitor a channel higher than the maximum you set it to. Make sense?

And, oh yeah, the STAC remembers the last set Tally Channel across power cycles. Convenient, yes?

Setting the Tally Channel (V-160HD)

Setting the Tally channel on the V-160HD is essentially the same as described above. Blue numbers represent HDMI channels. Pale green numbers are the SDI channels.



So to set the active tally channel, hold down the button until the channel select colors appear. Release and then click the button to pick the next channel. When the last HDMI channel is shown, clicking the button again will move to select the SDI channels as noted by the display select state colors changing to orange on a pale green background. When the last SDI channel is shown, the next click will wrap back to the first HDMI channel again indicated by orange numbers on a blue background.

Once the desired channel is shown, a long press will select that channel. The selection confirmation checkmark is shown and you're good to go.

Setting the Tally Display Mode

Tally Display mode? Yup.

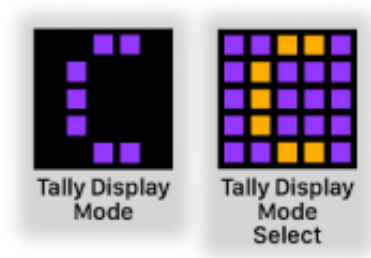
Folks on-stage (the "talent") really only need to know where the cameras are and which camera is currently live. For the person operating the camera it is also useful for them to know whether the channel is inactive (neither in program nor preview) and to be informed if there are any error conditions or operating issues with the STAC. To accommodate this, you can select the Tally Display Mode of the STAC to either "Camera Operator" or "Talent".

In Talent mode, the display shows red when the tally channel being monitored is live (a.k.a.: Active, Program, PGM, On Air) or green otherwise. Refer back to the picture [up there](#) in the [Quick Start](#) section.

In Camera Operator mode, the display is red when the channel is live, green when in Preview (a.k.a.: Inactive, Standby, PVW) and "dotted purple" when Unselected (neither in Program nor Preview). If the operation of the STAC becomes anything other than normal, the display changes to the alert the operator. More on that down in the [Troubleshooting](#) section.

The sequence of steps to change the Tally Display Mode is the same as for setting the tally channel so this section is abbreviated. Check out [Setting the Tally Channel](#) above for the long-winded version.

The Tally Display Mode is shown as a purple letter on a black background. A **C** means the STAC is in **Camera Operator Mode**. A **T** is shown for **Talent Mode**.



To change the Tally Display Mode:

1. When the purple **C** or **T** is shown, press and hold the button.
2. After about two seconds the display will change to an orange letter on a purple background.
3. Click the button to toggle between the two Tally Display Modes.
4. When the desired mode is shown on the display, press and hold the button.
5. After about two seconds the green confirmed checkmark will be shown briefly and the STAC will exit select mode. Release the button.
6. The new Tally Display Mode will be displayed as a purple letter on a black background.

You can press and hold again to change the Tally Display Mode or if you click the button, move on to displaying the next operating parameter.

When in a select state, the STAC has a timeout feature. If after entering the select state, no clicks of the display button are detected for about 30 seconds, the select state is cancelled and the display and Tally Display Mode will revert back to the value it had before entering the select state. From here the button can be clicked to advance to the next parameter or press and hold it to enter the select state again.

Note: When the STAC is in Talent Mode, the WiFi connect icons show only on power up or when the STAC is reset. If WiFi is lost after having a WiFi connection, the display will switch to the Preview (green) state until WiFi is reestablished. Once connected, normal display operation will resume. In Camera Operator Mode, the WiFi connect information is always shown.

The STAC also remembers the last set Tally Display Mode across power cycles. Genius.

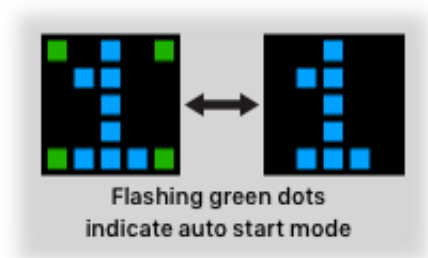
Setting the Startup Mode

Normally when the STAC starts up or is reset you have to "click through" the operating parameters to start the WiFi connect sequence for the STAC to begin to monitor the active channel. If you rarely change those parameters or if the STAC is installed in a place that is tricky to access, the STAC can be set to automatically jump to the WiFi connect sequence without the need to "click through".

This is the Startup Mode and it can be set to "Standard" or "Autostart".

In Standard mode you have to click through each operating parameter as described in [Quick Start](#), changing whichever one you would like along the way.

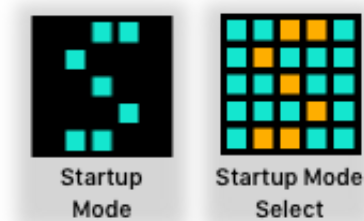
When started in Autostart mode, the STAC will show the active tally channel and blink the corners off and on in green to let you know autostart is active. The STAC will wait like this for about 15 seconds. If the button is not pressed during this time, autostart kicks in and the STAC will jump right to the WiFi connect sequence. If the connection is made, the STAC then starts to monitor the active tally channel.



If you need to make changes when in Autostart mode, just click the button. This cancels Autostart -- the corners stop flashing, and the active Tally Channel is shown. From here, click through to change the parameter you want and then on to the WiFi connect sequence.

The sequence of steps to change the Startup Mode is the same as for setting the tally channel so this section is short. Check out [Setting the Tally Channel](#) above for the long-winded version.

The Startup Mode is displayed as a teal letter on a black background. An **S** means the STAC is in **Standard Startup Mode**. An **A** is shown for **Autostart Mode**.



To change the Startup Mode:

1. When the teal **S** or **A** is shown, press and hold the button.
2. After about two seconds the display will change to an orange letter on a teal background.
3. Click the button to toggle between the two Startup Modes.
4. When the desired mode is shown on the display, press and hold the button.

5. After about two seconds the green confirmed checkmark will be shown briefly and the STAC will exit select state. Release the button.
6. The new Startup mode will be displayed as a teal letter on a black background.

You can press and hold again to change the Startup Mode or if you click the button, move on to displaying the next operating parameter.

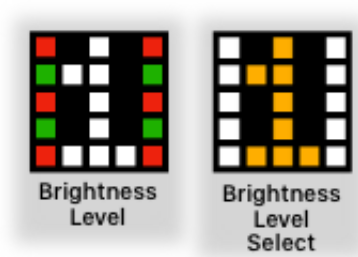
When in a select state, the STAC has a timeout feature. If after entering the select state, no clicks or presses of the button are detected for about 30 seconds, the select state is cancelled and the display and Startup Mode will revert back to the value it had before entering the select state. From here the button can be clicked to advance to the next parameter or press and hold it to enter the select state again.

The STAC also remembers the Startup Mode across power cycles. It's a wondrous thing.

Setting the Brightness Level

To accommodate a range of ambient light levels when using the STAC, the brightness of the display can be changed. As with changing the active tally channel, the method is the same so this section is abbreviated. Check out [Setting the Tally Channel](#) above for the extended directors cut.

The current brightness level is shown as a white number on a spectacular field of red and green dots. Yes, it looks like Christmas. There are six brightness levels, increasing from **1** (lowest) to **6** (brightest). Bet you didn't see that coming. Anyway, as levels are selected, the display brightness changes to match.



To change the Brightness Level:

1. When the current Brightness Level number is shown, press and hold the button.
2. After about two seconds the display will change to an orange number with a white stripe on either side.
3. Click the button to increase the Brightness Level.
4. At Brightness Level 6, the next click will wrap around to Brightness Level 1. You can keep

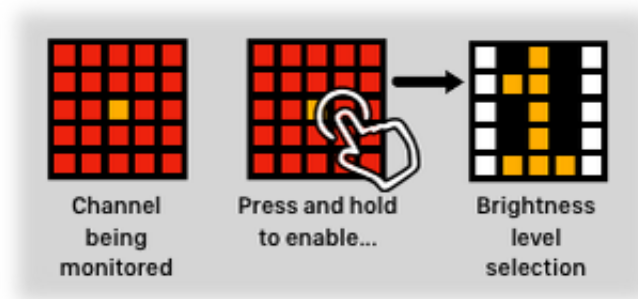
clicking to find the ideal photon bombardment.

5. At the desired Brightness Level, press and hold the button.
6. After about two seconds the green confirmed checkmark will be shown briefly and the STAC will exit the select state. Release the button.
7. The new Brightness Level is set.

You can press and hold again to change the Brightness Level or if you click the button, the STAC will start the WiFi connection sequence.

When in a select state, the STAC has a timeout feature. If after entering the select state, no clicks of the display button are detected for about 30 seconds, select state is cancelled and the display and Brightness Level will revert back to the value it had before entering the select state. From here the button can be clicked to advance to the WiFi connect sequence or press and hold it to enter the select state again.

Here's the bonus feature. *The Brightness Level can be changed while the STAC is monitoring the active tally channel!* Works in either Camera Operator or Talent mode. Mind blowing I know. The method is the same as described here. Just press and hold the display button. The STAC will flip into the Brightness Level select state. Set the level as you'd like, then press and hold the button. The confirmation checkmark will be briefly shown. Release the button and the STAC will return to monitoring the tally channel. Whoa! And yes, the timeout feature is active here as well.



The STAC engages its elephant brain to remember the last set Brightness Level across power cycles. One and done!

Setting the Display Orientation

As the world turns, it may not always be convenient to mount the STAC with the USB port at the bottom. It'd be nice if the display was always an upright citizen. Well, we got you covered there too.

On startup, the STAC determines which way is up and will rotate the display so that things are

always shown right side to. If the STAC is started laying on a flat surface, the USB port is considered down and that will be the baseline for the display.

To change the display orientation, mount or hold the STAC as it will be in use and then click the Reset button on the side. The STAC will restart and set the display orientation appropriately. Is that cool or what!

The STAC does this check on every power up or restart. Orientation from the last power up or restart is not retained.

Peripheral Mode

Often on video camera rigs, two STACs are used. One facing the camera operator and one facing the stage talent. The units act as an operator-talent pair. Peripheral Mode allows one STAC to receive its tally information via a cable from the other.

When set up like this, one STAC becomes the tally controller and the second becomes a tally peripheral to the controller. This reduces the traffic on the WiFi network and simplifies power cabling at the camera.

Enabling Peripheral Mode

You will need two STAC's.

First, remove all power from each STAC.

1. Connect the two STACs with a 4-pin GROVE cable Refer to the Notes below for cable details.
2. Connect one end of a USB-C cable to one of the STACs (either one will do) and the other end of the USB-C cable to a 5VDC power supply.

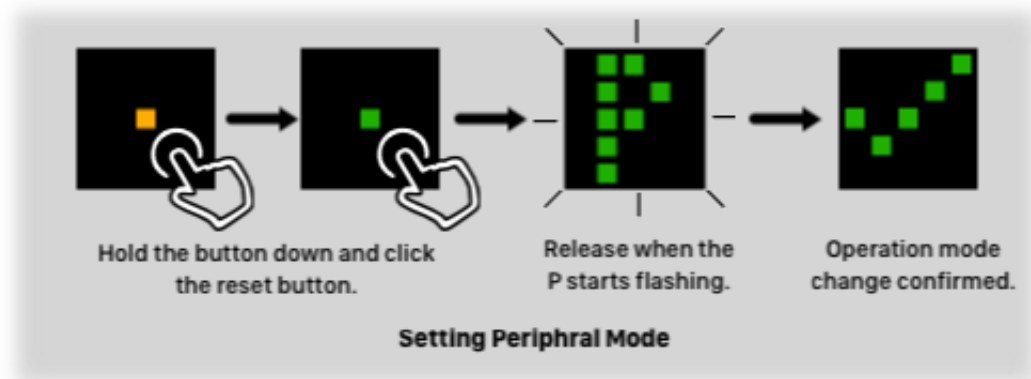
Only one of the STACs should be powered via its USB-C port. The other STAC is powered through the GROVE cable.

To put a STAC into Peripheral Mode, we need to do a bit of a dance with the side reset button and the display button. This sounds a bit tricky but you'll get the hang of it in no time. Best to read through the directions before giving it a go for the first time.

On the STAC you want to operate in Peripheral Mode:

1. Press and hold down the side reset button.

2. While continuing to hold down the reset button, press and hold down the display button.
3. While continuing to hold down the display button, release the reset button.
4. The centre display pixel will come on, quickly followed by it changing briefly to green.
5. A green letter "P" will then start flashing on the display.
6. As soon as you see the "P" flashing, release the display button.
7. The "P" will stop flashing and quickly thereafter a green checkmark will be shown.
 - The "P" indicates that Peripheral Mode operation has been selected, and is confirmed by the checkmark.
8. The STAC will then restart and begin operating in Peripheral Mode, receiving its tally information from the controller STAC.



Note

If you hold the button down longer than about two seconds after the "P" starts to flash, you will see other icons start to flash on the display, so it is important to release the button as soon as the flashing starts.

If do you see anything other than the "P", it's not the end of the world, just means you've discovered some other very cool features of the STAC that are covered down in the section on [Maintenance Functions](#).

Operation

On power up or restart, to alert you that the STAC is in Peripheral Mode, it will flash a green "P" on the display a few times before getting to work.

In Peripheral Mode, all normal operation state and features of the STAC are disabled. The STAC will not connect to a WiFi network, nor poll the Roland switch and it will ignore the normal Tally Channel, Tally Display Mode and Startup Mode settings. All tally information is received from the other (controller) STAC via the GROVE cable.

When in Talent mode, the display will be just as described [above](#). It will show red if the video channel being monitored by the controller STAC is in Program or onair and it will be green otherwise.

If set to Camera Operator, the display is as described [above](#) except that any state other than program, preview or unselected is shown as an orange X.

The display Brightness Level select feature is here too.

Setting brightness and tally display mode are a bit different when in Peripheral Mode. But not so much that it's a biggie. Either can be set any time after the green flashing P is shown.

Setting the Brightness Level

1. Press and hold the button. In about two seconds, the STAC moves to its Brightness Level select state with the current brightness level displayed just as described [above](#).
2. Release the button.
3. Now you can click the button to pick the new level.
4. Press and hold the button and the new level is confirmed by showing the green confirmation checkmark. Release the button and the STAC returns to monitor the tally status.

Setting the Tally Display Mode

Flipping between the two modes is done like so:

1. Press and hold the button. In about two seconds, the current Brightness Level will be displayed.
2. Keep holding the button down until the current Tally Display Mode is shown. The STAC will already be in select state.
3. Release the button.
4. Click the button to toggle between Talent and Camera Operator.
5. Press and hold the button and the new tally display mode is confirmed by showing the green confirmation checkmark. Release the button and the STAC returns to monitor the tally status.

See the bits above on setting the [Brightness Level](#) and [Tally Display Mode](#) for all the deets.

Disabling Peripheral Mode

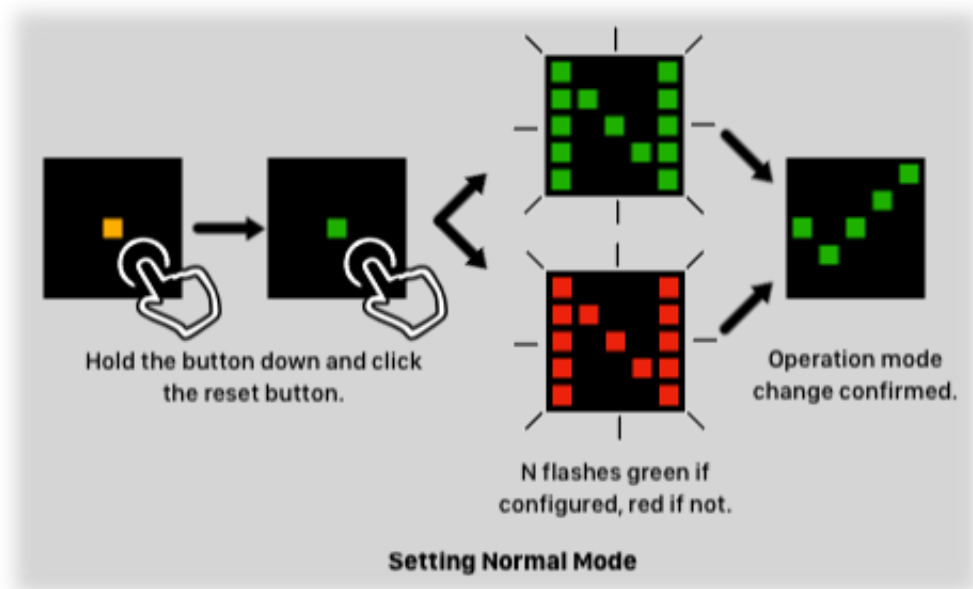
To disable Peripheral Mode and return the STAC it to its normal operating state:

1. Disconnect the USB cable powering the pair of STACs.
2. Remove the GROVE cable connecting the STACs.

Now we need to do again the little two button dance used to get the STAC into Peripheral Mode.

This sounds a bit tricky but you'll get the hang of it in no time. Best to read through the directions before giving it a go for the first time.

1. Connect a USB cable and power up the STAC that was operating in Peripheral Mode.
2. Press and hold down the side reset button.
3. While continuing to hold down the reset button, press and hold down the display button.
4. While continuing to hold down the display button, release the reset button.
5. The centre display pixel will turn orange, quickly followed by it changing briefly to green.
6. A letter "N" will then start flashing on the display.
7. As soon as you see the "N" flashing, release the display button.
8. The "N" will stop flashing and quickly thereafter a green checkmark will be shown.
 - The "N" indicates that Normal Mode operation has been selected, and is confirmed by the checkmark.
 - If the STAC was configured before being placed in Peripheral Mode, the "N" will flash green and it will return to Normal Mode operation after restart.
 - If the "N" flashes red, it's a heads up that the STAC has never been configured and you'll will need to do that before it will operate in Normal Mode.
9. The STAC will then restart.



1. If you hold the button down longer than about two seconds after the "N" starts to flash, you will see other icons start to flash on the display, so it is important to release the button as soon as the flashing starts.
2. If do you see anything other than the "N", it's not the end of the world, just means you've discovered some other very cool features of the STAC that are covered down in the section on [Maintenance Functions](#).
3. It is not required to configure the STAC before Peripheral Mode can be used.
4. Peripheral Mode retains its own display brightness level and tally display mode settings independent of those for its normal operating state. The values are retained until changed (when in Peripheral Mode) or when a Factory Reset is performed—in which case the Peripheral Mode display brightness will be set to **1** and the tally display mode to Talent.
5. The orientation feature works in Peripheral Mode so the display will always be shown the right way up. Pop back up to [that section](#) to review.

Hardware Implementation

Connection

The cable needed to connect the two STACs is known as a GROVE cable, as popularized by [seeed Studio](#). More specifically, the cable is a female-female 4-pin HY series, 2.0mm pitch.

Searching for a 4-pin GROVE cable should turn up a few sources.

Alternately, if searching for an HY-2.0 4-pin cable, make sure to purchase one with "standard" pinout where pin 1 is connected to pin 1, 2 to 2, 3 to 3 and 4 to 4. **Some companies offer a "reversed" cable** (pin 1 to pin 4, 2 to 3, 3 to 2, 4 to 1) **which will irreparably damage the STAC if used**. So, like, don't use it.

Either "buckled" or "unbuckled" connectors on the cable work. Buckled connectors have a latch that will lock the cable connectors to the STAC.

Be aware that the ATOM Matrix GPIO pins use 3.3V logic levels and are not 5V tolerant. Driving any GPIO pin beyond 3.3V will irreparably damage the device.

In Peripheral Mode

In Peripheral Mode, the STAC continually monitors the state of G26 and G32 on its GROVE port and uses the state of those inputs to determine the state of its tally display as per the table below.

G26 (TS_1)	G32 (TS_0)	Displayed Tally State	
		Talent	Camera Operator
0	0	PVW	X
0	1	PVW	UNSELECTED
1	0	PVW	PVW
1	1	PGM	PGM

Where:

1 = logic high (true)

0 = logic low (false)

PVW (Preview, Standby, Selected) = GREEN

PGM (Program, Live, On Air) = RED

UNSELECTED = "PURPLE DOTTED"

X (Unknown) = Orange X

When operating in Peripheral Mode, the GPIO pins are configured as input with the internal pull-down resistors enabled. The pull-down configuration ensures that when no external device is driving the pins, they default to the unknown state.

Here's the cool part. This allows the opportunity for developers to use a STAC set in Peripheral Mode as a tally indicator driven by an external piece of gear. Simply connect a cable to the GROVE port, apply power and drive pins G26 and G32 as needed.

In Normal Mode

A STAC running in its normal operating state (not in Peripheral Mode) acts as a controller by outputting the current tally state of the video channel it is monitoring to its GROVE connector. Pins G26 and G32 on this port are set as per the table below.

G26 (TS_1)	G32 (TS_0)	ROLAND SWITCH TALLY STATE
0	0	UNKNOWN
0	1	UNSELECTED
1	0	PVW
1	1	PGM

Where:

1 = logic high (true)

0 = logic low (false)

UNKNOWN = the STAC is not connected to the Roland switch; or no information was reported from the last poll; or the return from the last poll was not an expected response; or the STAC is in an error state.

UNSELECTED = the video channel being monitored is neither in PVW or PGM.

PVW = the channel being monitored is in preview (Preview, Standby, Selected)

PGM = the channel being monitored is in program (Program, Live, On Air)

When operating in Normal Mode, the GPIO pins are configured as output with no internal pull-up or pull-down resistors enabled.

This allows the opportunity for developers to use a STAC running in its normal state to provide tally information to another piece of gear.

To do this, connect a cable to the GROVE port, power up the STAC and monitor the state of pins G26 and G32.

This could allow the STAC to:

- drive a larger or brighter indicator;
- act as a wireless tally relay;
- be integrated as an indicator and controller into a system such as [Tally Arbiter](#).

The "UNSELECTED" state is useful as it indicates normal operation and communication of the controller STAC with the Roland (or emulated) device being monitored even if that channel is not active in either PGM or PVW.

With proper cabling, multiple STACs operating in Peripheral Mode could be connected in parallel to a STAC operating in its normal state. No testing has been done to confirm how many STACs

could be driven in this manner. Power supply would be an issue to consider as well.

Backwards Compatibility

Peripheral Mode has been available since STAC software version 1.1.0. Using Peripheral Mode required installing a jumper wire between two connection points on the back of the ATOM.

Starting with STAC software version 3.0.0, a jumper is no longer required nor supported. STAC devices running version 3.0.0 or greater should remove the jumper wire and use the methods above to enable and disable Peripheral Mode.

That said, a STAC running software version 1.1.0 to 2.2.1 using a jumper to enable Peripheral Mode, will work in Peripheral Mode with a STAC running software version 3.0.0 or greater.

Likewise, a STAC running version 1.1.0 to 2.2.1 as a tally controller, can be drive a STAC running software version 3.0.0 or greater that is operating in Peripheral Mode.

Settings Deep Dive

This is excitin'! When you first went through the [First Time Setup](#) you certainly noticed a number of other widgets on the Setup page. We're diving into what those are all about here. Take a deep breath and jump in.

If you are setting up multiple STACs the Quick Setup area of offers two things to speed this up and prevent entry errors.

If you click the blue `Choose File` button, a dialog will pop up that'll let you navigate around your device and pick a STAC configuration file. Once you've done that, another dialog box will pop up letting you know the setup from that file has been loaded into the setup form (you'll see those settings after you've clicked the `OK` button).

Now you can scroll down the form and click the `Configure STAC` button just as if you entered everything manually. Boom and no typo's.

"But how do I get a STAC configuration file!" you ask? Well...

After you have manually entered all the information into the setup form, click on the blue `Save Settings` button. This will create and save the information in the form to, you guessed it, a STAC configuration file on your device. (Where exactly that file is stored depends on the device you're using.)

The file name is:

- `stac-v-60hd-config.json` for a Roland model V-60HD or,
- `stac-v-160hd-config.json` for a Roland model V-160HD.

These are text files in standard JSON format that can be moved, copied, shared and backed up as you would like.

What is even better, when the file is created, it only copies information that is entered in the form when you saved it. So, if you don't want for example the SSID password saved with the file, leave that field empty, fill in the rest and the save. When you load that file, fill in the blank fields manually and complete the set up.

The resulting configuration file can be renamed to what you'd like. If you know your way around the JSON syntax, you can edit the contents of the file as well.

But wait, there's more!

What if you want to configure multiple STACs but don't want to save a file? Only easy. Fill in the form as needed and then click the green `Copy Settings` button. This places the information in the form onto the clipboard of your device. Once you've configured that STAC, connect to the hot spot of the next one and then click the green `Paste Settings` button in the Quick Setup area at the top of the Setup tab. If you're using a mobile device the browser should show another `Paste` button you'll need to click to copy the information into the form. If you're using a desktop machine, a dialog box will pop up telling you click the `OK` button then use the keyboard paste command shortcut to fill the information into the form. Those are the browsers security measures kicking in to make sure you know you're pasting into the web page.

When loading a configuration file or pasting into the form it is not necessary to pick the model of the switch first. But if you have, and then paste or load a configuration that is different from the model chosen, the STAC will show a dialog letting you know that the chosen model doesn't match, asking if you want to proceed. Clicking `OK` will load the information and present the completed form for you to review before sending it to the STAC.

If the selected settings file or on the information on the clipboard is not STAC configuration information, a dialog box will pop up letting you know that. Redo the file selection and you should be all set.

If you've already configured a STAC and need to reconfigure it or want to save the information in a configured STAC as a template for others, you'll need to fire up the STAC hotspot to do so. It's pretty easy and covered next in [Maintenance Functions](#).

////////////////////////////////////

Maintenance Functions

Ah change. The thing we all love. Or not. But required sometimes nonetheless. Welcome to the STAC maintenance functions.

These functions allow you to; change things up, keep the STAC running in tip-top shape, probe the depths of its brain, and wipe its mind.

The maintenance functions are:

1. Redo the setup;
2. Update the STAC software;
3. Perform a factory reset (which can also be done directly on the STAC).
4. View and copy information about the STAC setup and software;

Getting to the Maintenance Functions

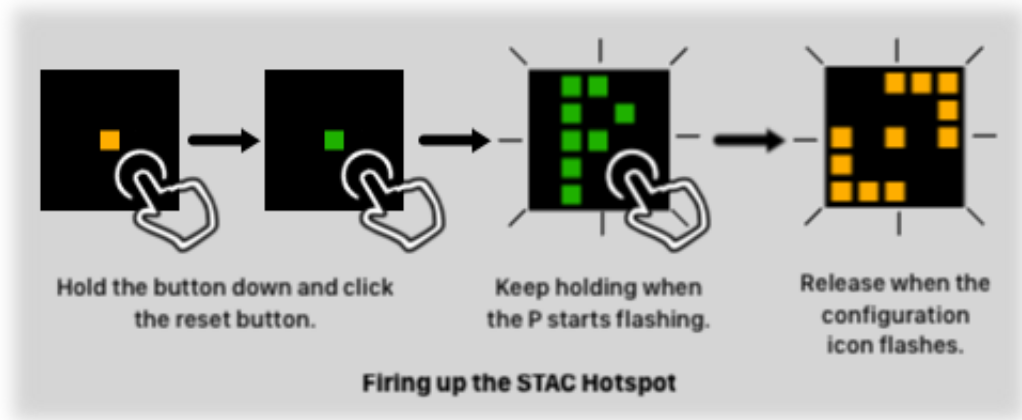
To do maintenance things, we need to fire up the STAC hot spot and then connect to it with a web browser.

To do that, we're going to do a little dance with the reset and display buttons in combination. If this is your first maintenance malagueña, best to read through the steps below first and then give it a shot.

1. Power up the STAC. It will either pause at the tally channel display (if autostart is active just click the button to cancel it) or do the flashy "P" thing if its set to Peripheral Mode
2. Press and hold down the Reset button on the side of the STAC.
3. While continuing to hold down the Reset button, press and hold down the Display button.
4. While still holding down the Display button, release the Reset button. (Insert gratuitous Kraken reference.)
5. Keep holding down the Display button until the "N" or "P" stops flashing and then...
6. Release the Display button when the orange setup icon starts flashing on the display.

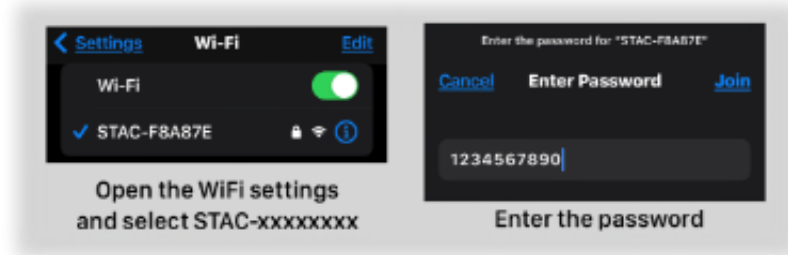
If you did this little dance right, (guess you figured it'll take two hands) the orange setup icon will stop flashing and then start to pulse in brightness. The STAC is now waiting patiently for you to take action.

If the STAC returns to the tally display (or flashy "P" thing), just give the two-button dance another whirl. You got this.



7. If the STAC has never been configured, after powering on, the configuration icon will flash red then start to pulse in brightness and its WiFi hotspot will be fired up already.
8. Now grab your favourite device, open its WiFi settings and pick the SSID of your STAC. The STAC SSID starts with "**STAC-**" followed by eight characters. Something like "**STAC-4660A124**".

Each STAC has its own unique SSID.



9. Join the STAC WiFi hotspot and when prompted, enter the STAC WiFi password which is **1234567890**.
10. Once connected—on your smart device: fire up a web browser; open a new browser window and go the STAC setup page at `http://stac.local`.
11. Click the **Maintenance** tab and marvel at the cornucopia of invigorating options.

STAC Setup

Device: M5Stack ATOM Matrix
SSID: STAC-00F8A8
MAC: 94:89:7E:A8:F8:00
Version: 3.0.0-RC.25 (a4f031)

SetupMaintenance

Maintenance

Firmware Update

Instructions:

1. Select a .bin firmware file below
2. Click "Update Firmware"
3. Wait for upload and installation to complete
4. STAC will restart automatically

Note: To cancel, press the STAC reset button.

Select Firmware File (.bin):

Choose FileNo file chosen

Update Firmware

Factory Reset

Perform Factory Reset

Show Info

STAC Maintenance tab

We're going to walk through each in detail below.

Notes

1. If you hold the button down longer than about two seconds after the Configuration change icon starts to flash, another icon starts to flash on the display, so it is important to release the button as soon as the flashing starts.
2. If do you see anything other than the Configuration icon, it's not the end of the world,

just means you're ahead of the class and have done a Factory Reset which is covered [later](#).

3. If you're having trouble connecting to the STAC using `http://stac.local`, it may be because the `http:` bit wasn't entered and so the browser is trying to use an `https:` connection. If checking that doesn't work, you can always connect by pointing the browser to the IP address of the STAC which is `192.168.6.14`.
4. If you change your mind at this point, click the Reset button and the STAC will shut down its hotspot and return to its normal startup sequence.

Redoing the Setup

At some point, probably, one or more of the WiFi SSID, Password or Roland device IP address will change. Or maybe you want to take your STAC on the road. Who doesn't like a road trip? In this case you'll want to set up the STAC to match the new settings. We got you covered.

Follow the steps in [Getting to the Maintenance Functions](#) to get to the maintenance functions, except click the Setup tab, not Maintenance.

From here follow the directions under [First Time Setup](#) above; connecting to the STAC by WiFi and using a browser to bring up the Setup forms and so on.

If you mysteriously entered this setup state by accident or you change your mind at this point, click the Reset button and the STAC will return to its normal startup sequence.

That's all there is to it.

"One more thing". If a STAC was configured for one model, and is later configured for a different one, the tally channel, tally operating mode, startup mode and display brightness level are all reset to their defaults. If the model is not changed, these settings are retained.

Updating the Software

This is too cool. The STAC software can be updated using pretty much any device that can connect to WiFi and run a web browser. This is called an "over the air" or OTA update. Mind blown. Again.

OK. In the repository where you found all this stuff in the first place, inside the STAC folder is a folder called `bin`. Inside there is a file named something like "`STAC_v3.0.0-
_AtomMatrix.bin`". This file contains the STAC software. Our objective is to download that file to the STAC.

We do that by:

- Copying the " `STAC_XXXXXX.bin` " file to the device you'll use to talk to the STAC. Same type of device you used to do the setup.
- Connecting to the STAC.
- Choosing the " `STAC_XXXXXX.bin` " file on your device.
- Sending that file over to the STAC.

All right. Let's get into the steps, each in all their incredible glory.

1. **Copy the file to your device**

Actually, not a lot to say here. If you're using a computer computer, it should already be there if you downloaded the STAC repository. If you're using a mobile device, it's "an exercise left to the reader" to determine how to copy the file to your device. Too many permutations to cover here.

2. **Connect to the STAC**

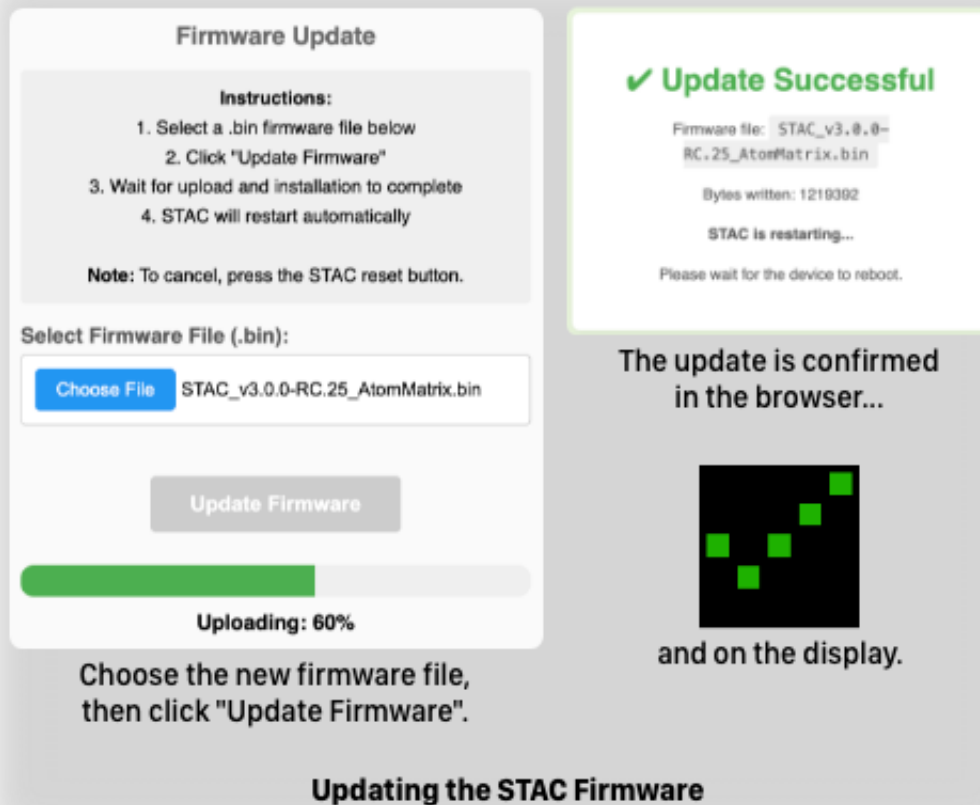
Follow the steps in [Getting to the Maintenance Functions](#) to get to the maintenance functions.

3. **Choose the file**

Click the blue `Choose File` button and navigate to the location on your device where you saved the file and select it.

4. **Update the firmware**

Tap the `Update Firmware` button. The progress bar should start to move. In about ten or so seconds, the browser page will refresh to let you know the status of the update. The STAC display will briefly show a green checkmark.



After updating its software the STAC will shut down its hotspot and then restart, returning to its normal startup sequence fresh as a daisy with new clean software.

If the software update fails, the STAC will show a red X and the browser will a dialog box with a reason for the failure. Gross. Some things to try in this case are, make sure:

- the correct " STAC_XXXXXX.bin " file was selected;
- you're not so far away from the STAC that the WiFi connection was lost;
- the power connection to the STAC is good.

Try the update again. If things are still wacky, open a Discussion topic on the [STAC GitHub site](#) including a screen shot of the error message. We'll try and sort things out.

That's it. Super easy indeed!

Other things to know about updating

Here are a few points worth noting when doing a software update.

1. After a software update, the STAC will try and use the setup information it had before the update. However, depending on the nature of the update, it may be necessary to reset this information. If that is the case, when the STAC restarts, it will flash the red setup required icon on the display letting you know a new setup is required. The Release Notes will let you know if this is the case.
2. If the STAC is running software version 2.0 or lower, you won't be able to update the software as just described. You'll have to update it to at least version 3.0.0 by flashing the full version binary to the ATOM. But afterward you're golden!
3. If the STAC is running a version of software that required a jumper to enable Peripheral Mode, you will need to remove the jumper before a software update can be performed.
4. If the STAC has never been configured, or if a factory reset was done beforehand, when you go to put the STAC into software update mode the STAC software update icon will immediately start to flash. No need to setup or factory reset a STAC that is already a blank slate.
5. The software update can be cancelled any time before tapping the Update button in the browser by clicking the STAC Reset button. No harm, no foul.
6. If the STAC is reset or loses power or the WiFi connection before the update completes, the STAC should return to the same state it was in before the update was tried. Give the update another go.

Factory Reset

All righty then. Maybe you want to hand off your STAC to someone else or just for giggles restore the STAC to its "fresh-out-of-the-box" state. Doing this literally wipes all configuration information from the internal memory of the STAC: the switch SSID, password, Roland Smart Tally IP address, port number, maximum tally channel number and polling interval, all gone.

After a factory reset, you can remove power or, to put the STAC back into service, click the reset button and then set things up like [First Time Configuration](#) above.

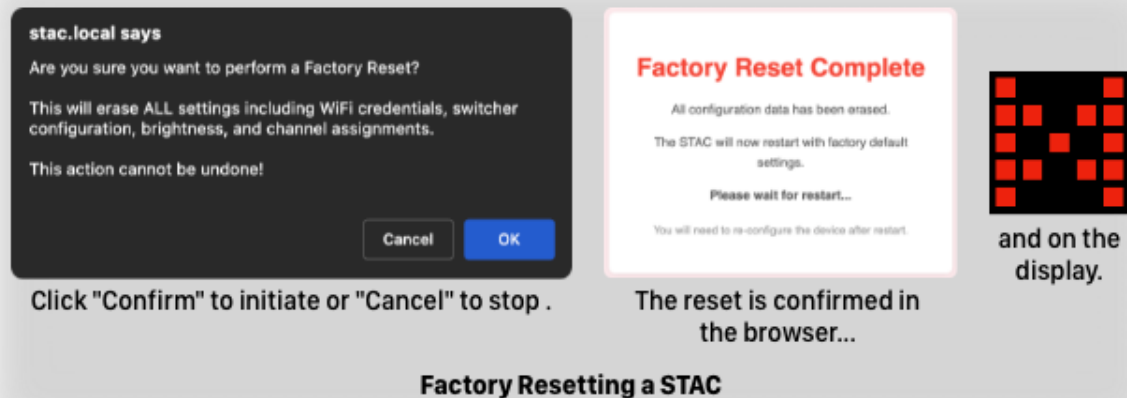
A factory reset can be done using a browser or directly on the STAC. Let's cover doing it with a browser first.

Using a Browser

Follow the steps in [Getting to the Maintenance Functions](#) to get to the maintenance functions.

Scroll down to the big red button that not so subtly says `Perform Factory Reset`. Clicking that will show a dialog box warning that you're about to build an intergalactic highway through the orbital path of the Earth so time to say good-bye to all your family and friends. Just seeing if you're paying attention. So no, not quite that dramatic, but it will warn you that you're about to wipe the mind of the STAC and it gives you an option to cancel.

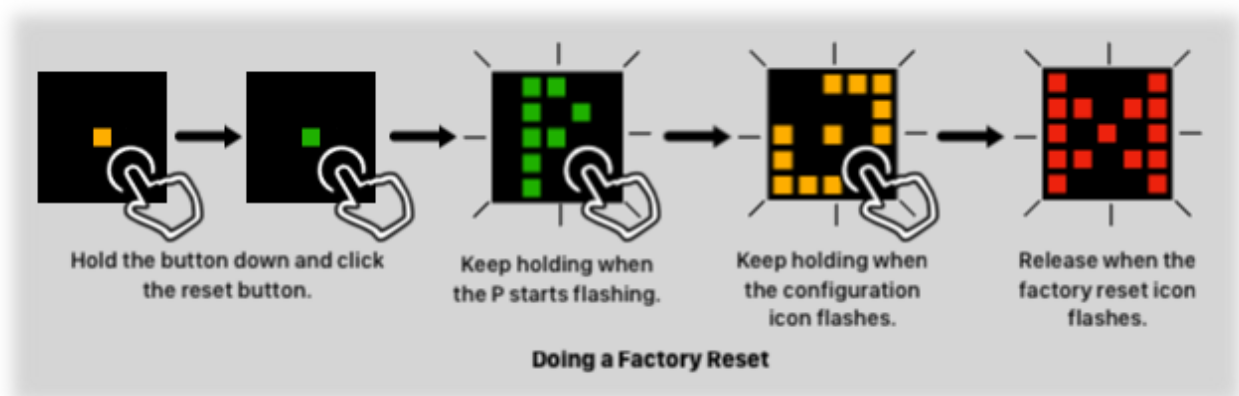
Clicking OK, will zap all the settings.



The STAC display will show the weird "double bar X" factory reset icon on the screen, blink once when the reset is complete and then pause. Disconnect power to the STAC and you're done. Or click the reset button if you're in the mood to reconfigure it as in [First Time Configuration](#) above.

From the STAC

To do a factory reset, using the buttons on the STAC follow the steps in [Getting to the Maintenance Functions](#) except, *keep holding down the display button* until you see the factory reset icon flashing on the screen.



Release the button. A moment later the STAC will be completely reset, the display will blink once to confirm and the factory reset icon will remain on the display.

Using the buttons, once the factory reset starts, there is no turning back. Not a super big deal, just do the setup thing once more and you're good to go.

Using the STAC buttons, you cannot do a Factory Reset if the red [Configuration Required](#) icon is displayed as the STAC is already in its factory default state.

Deep Secrets

And last but certainly not the least of the maintenance functions is the Show Info button at the bottom of the tab.

Not much more to say other than, click the Get Info button and up pops a window that gives a bunch of, well, information about the settings and some techie stuff about the STAC.



STAC Information Detail

All of it is pretty self explanatory, except maybe the three lines at the bottom which are details about the development environment used to build the STAC software. If you're really curious about what that means, drop us a line, but it's one of the things that's useful if you need to report a bug, in which case...

Clicking the button puts all that information on the clipboard of the device you're using to talk to the STAC with. It can then be saved to a file for documenting your tally light system or framed and hung on the wall. But that might be a bit odd perhaps.

To dismiss the window, click the button.

Back to the Beginning

One more thing...

The sections above show how the Reset button is used with the Display button to set up or factory reset the STAC.

The other trick is, when used by itself, the Reset button will return the STAC to its startup sequence. Why would you want to do this? Most common reason is let's say you just confirmed the tally channel, operating or startup modes but then realized you actually meant to change one or the other. Or maybe the USB power cable is in the way of something on the rig and you'd like to rotate the STAC to fix that up.

In this case, click the Reset button. The STAC display will go blank, the power light will come on, the STAC will determine which way is up, followed by the display of the active tally channel. Now you can press and hold to change that or click through to the tally display and startup parameters and put those where you want them. The brightness level can be changed while the STAC is operating so no need to restart the STAC in that case, but you could click the Reset button to get there too.

Much easier than removing and reconnecting power.

This holds true when the STAC is in Peripheral Mode as well, though the only reason to do this would be to re-orient the "up" position of the STAC so the brightness level numbers and tally mode characters are rotated as you wish.

Special Configurations

In most situations the STAC is used to communicate directly with a physical Roland switch or

device. When this is the case the IP address and port number entered on the web browser configuration form are the those of that Roland device. Just as described in [First Time Setup](#).

However, if you are using a STAC in an environment where the STAC is instead communicating with some other device or service that emulates a Roland Smart Tally switch, it may be required to change the STAC Smart Tally IP address and port number so that the STAC and that system are on the same wavelength. An example would be using the STAC with [Tally Arbiter](#) where tally devices from a variety of providers can be aggregated into a unified tally information system.

Refer to the documentation from the provider of the Roland emulator or service on how to configure the STAC if this applies to you.

When using a STAC with Tally Arbiter, the method used to identify a specific STAC is via the STAC tally channel number. The maximum tally channel that can be configured on a STAC is **8**. Thus no more than eight STACs can be used in a Tally Arbiter system. Unless a STAC is configured in Peripheral Mode, in which case the maximum number of STACs would be **16** with the understanding that a STAC operating in Peripheral Mode is not receiving independent tally information from Tally Arbiter.

Hey, we kind of got our geek on in here. If this entire section is sounding completely foreign, we get it. Just ensure when you configure the STAC that the IP address matches the actual Roland device and that the port is set to its default of **80** and you'll be in business.

For even more geeky stuff, check out the *Peripheral Mode Application Note.md* file in the repository as well.

Troubleshooting

Ahh life. That thing that happens when you're busy making other plans. Believe it or not at times the STAC may get uppity and display a couple of things to let you know it's not quite happy.

Fortunately, the list of things that can go wrong is short. (Oh boy, just challenged Murphy to a duel!)

The STAC is pretty robust and it will usually correct itself after encountering an error, returning to its normal operating state in short order.

If something is amiss, take a look below at the symptom and a step or two you can take to fix 'er up will be there. If not, shoot us a note and we'll see what we can do.

But before charging forth, know that the STAC will mask most errors on the display when its Tally Display Mode is set to Talent. No need to confuse the on-stage folks.

By the bye, when in Peripheral Mode, all errors are shown the same. Scroll back [there](#) for info.

The WiFi Connect Loop of Death

What you're seeing:

The STAC display shows the orange connecting to WiFi icon and then after a minute it changes to a flashing red WiFi icon and then back to orange.

This condition may be seen in Camera Operator mode and when first powered up in Talent mode.

What's causing it:

The STAC is trying to connect to the WiFi network. It tries this for about a minute and if it can't it flashes the red icon to let you know it's trying but there is still no connection. It keeps trying forever.

What to try:

- *If the STAC has successfully connected to this WiFi network before:*
 - Check that the WiFi access point is turned on.
 - Check that the STAC is within range of the access point. Move it nearer to the access point and see if it can connect.
 - Has the SSID or Password of the network been changed? Check with someone that might know. If so, whew—problem solved. Just pop up to the section on [Redoing the Setup](#). Check and re-enter the new SSID and Password information, submit that and the other info required and you'll be great.
- *If the STAC has never connected to this network:*
 - It is probably misconfigured. Likely means one or both of the WiFi SSID or Password was entered incorrectly on the Setup form. Hey, it happens. We're all human. Just pop up to the section on [Redoing the Setup](#). Check and re-enter the SSID and Password, submit that and the other info required and you'll be great.

Tally Status not Changing

What you're seeing:

The tally status display on the STAC does not change. It's stuck on green, red or "dotted purple" (if in Camera Operator mode).

What's causing it:

Most likely the STAC is set to monitor a channel on the switch that is not changing.

What to try:

- Confirm with the person doing the switching that the correct tally channel is being monitored.
- If not, click the Reset button on the side of the STAC and when the tally channel is displayed, follow the steps under [Setting the Tally Channel](#) to change the active channel.
- If operating in Peripheral Mode and the controller STAC display is changing but the peripheral STAC is not, it is most likely a problem with the cable connecting the two units.
 - Confirm the cable is a "standard" GROVE cable.
 - Try re-seating the ends of the cable in both STACs.
 - Try a different cable.

A Big Orange X?

What you're seeing:

There is is a big orange X on the display!



What's causing it:

Only seen when the Tally Display Mode is set to Camera Operator, this is the STAC letting you know that it cannot communicate with the Roland switch.

Everything else is OK though.

What to try:

- *If the STAC has been working with this switch before:*
 - Check that the Roland switch is turned on.
 - Check that the switch is connected to the WiFi network.
 - Has the IP address or port number of the switch changed? Check with someone that might know. If so, whew—problem solved. Just pop up to the section on [Redoing the Setup](#). Check and re-enter the IP Address and port number of the switch, submit that and the other info required and you'll be great.

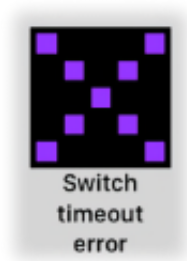
- *If the STAC has never connected to this switch before:*
 - The IP address or port number of the switch was probably entered incorrectly on the web browser Setup form. Pop up to the section on [Redoing the Setup](#). Check and re-enter the IP Address and port number of the switch, along with the other info required and you'll be good to go.

- *If the STAC is being used in an emulated environment:*
 - The IP address or port number of the STAC and the emulated device are probably not in sync. Check the configuration requirements of the system and change either the configuration of the system or the STAC as in [Redoing the Setup](#) so they match. See also [Special Configurations](#).

A Big Purple X?

What you're seeing:

There is is a big purple X (BPX) on the display!



What's causing it:

Only seen when the Tally Display Mode is set to Camera Operator, this is the STAC letting you know that it can connect with the Roland switch, but no replies are being returned when it is polled.

Everything else is OK though.

What to try:

- After asking the switch for the status of the active channel, the STAC waits about a second for the reply to come back. If it doesn't, the reply has "timed out". If the reply times out eight times in a row, the BPX is displayed.
- The STAC is continually polling the switch and next time round, the reply may come back before the time-out period. In this case the tally status of the active channel is shown, otherwise the BPX remains.
- If the BPX appears often but the STAC soon returns to displaying the active channel status, it may mean:
 - there are too many devices (STAC or other Smart Tally clients) asking the switch for status, causing a backlog;
 - the network is busy with other traffic (live streaming being a big one) causing delays in the communication between the switch and the STAC.

If this seems to be the case, have a chat with the some folks about the state of the network.

The Red WiFi Interrupt Screen!

What you're seeing:

The STAC was operating normally but suddenly a flashing red WiFi icon appeared on the

display, followed by the orange WiFi icon.

This condition is only seen when the Tally Display Mode is set to Camera Operator.

What's causing it:

The STAC was doing its thing but lost its WiFi connection. Bummer. The STAC is now in its WiFi connect sequence and will keep trying to connect.

If the WiFi connection is re-established, the STAC will automatically resume normal operation, displaying the tally status of the active channel. No need to click through the tally channel, operating mode and brightness confirmation steps or wait for autostart to finish.

Re-establishing the WiFi connection and getting everything back in sync with the Roland device may take a bit of time. But it will get there.

What to try:

If the STAC cannot re-connect to the WiFi network, Check for the same things under [The WiFi Connect Loop of Death](#).

Note:

When the Tally Display Mode is set to Talent and WiFi is lost after having a good connection on power up or restart, the display switches to the Preview state and remains there until the WiFi connection is re-established and resuming normal operation resumes.

Purple Question Mark?

What you're seeing:

There is a purple question mark on the display. It may appear intermittently or for longer periods of time.

This is only seen when the Tally Display Mode is set to Camera Operator.



What's causing it:

Ahh, well. What we have here is a failure to communicate. In a way that the STAC understands. Meaning the STAC sent a tally status request to the switch, a reply was received before the time-out period, but the reply was gibberish. If eight gibberish replies in a row are received, the purple question mark is displayed.

What to try:

Probably not a lot on this one. The STAC and the switch should get things sorted the next time they talk. If not, it's possible there are network congestion issues like mentioned in the [Big Purple X?](#) problem above.

If the STAC is being used in an emulated environment as in [Special Configurations](#) it's possible the emulated device is not responding in a way the STAC is expecting. If you think this is the case, take a look at the *STAC Communications.md* document in the repository.

The Big Red X

What you're seeing:

Well boss, there is a big red X on the display!



What's causing it:

Ack. Really hope you never see this one. It means the STAC had a big internal burp and got pretty confused.

What to try:

With this one there is really nothing to do. The STAC should sort itself out, recover and carry on with its life as normal.

If it does seem to be mis-behaving for a longer period of time, click the Reset button. If not set to autostart, you'll have to click through the normal startup tally channel, operating mode and brightness level displays after which the STAC should follow through with the rest of its normal WiFi connect startup sequence and then resume normal operation.

Acknowledgements

Exactly half an ohnosecond into our very first church livestream it became painfully obvious that radio coms alone weren't going to cut it. Our camera operators *needed* a tally light system. The search began.

Wired systems were plentiful but running extra cable about the building and tethering the cameras down didn't seem the way to go.

A bit of reading showed our newly acquired Roland V-60HD video switch had this funky Smart Tally capability. Surely I figured, that's the solution. We gave that a shot for a few weeks but having our operators use their own phones wasn't working out so well. Plus the displays were too large and the phones had to be configured apart from their normal setup each time to work properly. Back to the drawing board.

A few WiFi capable commercial options were found but the per unit cost was stupid expensive. And I mean waaaaay stewpid for what they did.

More searching led us to the [Tally Arbiter](#) project headed by the amazing Joseph Adams and the [Tech Ministry](#) folks. I set up Tally Arbiter with help from Joseph and with a bit of time searching and poking about figured how get the Tally Arbiter code running on an [M5Stack ATOM](#) and our livestream computer. It worked great and the size and low cost of the ATOM units meant it could be easily deployed.

However, we were only running with that single video switch and with the entire production and tech team being volunteers where only a few had a technical background, I was forced to come to the conclusion that as great as Tally Arbiter is, it was beyond the skill set of our team to set up, configure and maintain in a sustainable fashion.

But we had a dedicated WiFi network, these super little ATOM units and the Roland switch that could speak Smart Tally—so why not just let the two of them talk to each other directly?

There's a phrase about fools rushing in where angels fear to tread. And to not prolong things too much, I jumped in and started to bang my fingers on the keyboard and my head against the wall. With help from newly found on-line friends the first version of what is now STAC was hacked together in a few weeks. Add a dash of feedback from our teams and the feature list and capabilities grew to what is being released today.

So thanks again [Joseph](#).

For the support from the folks that hang around the [reddit Arduino](#), [Arduino Projects](#) and [ESP32](#) subs—I'm grateful.

To the awesome and dedicated folks on our production, tech and worship teams—applause and praise.

To those that have signed on to become contributors; I am humbled that you find this work to be of use. It's incredible to be able to learn as you share your knowledge and experiences.

Shout out to GitHub user `feta00` who first asked about the V-160HD and then went way above and beyond in support of the development to add that model. You rock.

But most of all to my wife who, as companion to a guy who can become obsessively dogged in pursuit of a goal, has shown—once more—saintly patience and forbearance as I chewed up an inordinate number of hours, days, and weeks to bring this project to fruition. My dear, you are a true angel and I remain your blessed fool. Thank-you.

- Xylopyrographer
April, 2024

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Revision History

2026-01-31:

- Revise for STAC software version 3.0.0.
- Significant revision with many parts of the document changed to reflect additional functionality and changes to features.

2024-05-09:

- Revise for STAC software version 2.2.0 which brings V-160HD support.
- Revise *First Time Setup* section to include new setup method.
- Revise *Setting the Tally Channel* section to *Setting the Tally Channel (V-60HD)* for V-60HD items.
- Add *Setting the Tally Channel (V-160HD)* section for V-160HD specific items.
- Revise *Redoing the Setup* section to include V-160HD items.
- Revise *Factory Reset* section to include V-160HD items.
- Usual grammatical and spelling touch-ups.

2023-02-03:

- Revise for STAC software version 2.1.
- Revise *First Time Setup* section
- Revise *Peripheral Mode* section
- Add *Updating the Software* section
- Usual grammatical and spelling touch-ups.

2022-08-09:

- Revise for STAC software version 2.0.
- Add note about IP addressing under *First Time Configuration*.
- Add "Orange X" error description section.
- Add "Polling Interval" configuration option.
- Revise configuration screen images.

2022-01-04: Revise for STAC software version 1.10, detailing Peripheral Mode operation.

2021-04-30:

- Revise for STAC software version 1.9.
- Add Display Orientation.
- Add Contents.

2021-04-15: Revise for STAC software version 1.8. Add Startup Mode.

2021-04-05:

- Add user configurable Smart Tally device port number.

- Add "Special Configurations" section.
- Correct STAC response to losing WiFi connection when operating in "Troubleshooting".
- Other minor changes.

2021-04-01: Change all pictures to use the `` syntax. Fixed a formatting error.

2021-03-31: Add the "Back to the Beginning" section. Other minor editorial changes.

2021-03-29: First release.

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